

The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern and dynamic visual effect.

Lecture 8

Web Development

Objectives

- ▶ Web Development
- ▶ HTML/ Basic HTML tags
- ▶ Headings
- ▶ Title
- ▶ Paragraph
- ▶ Tables
- ▶ Forms
- ▶ Images
- ▶ Links
- ▶ Lists
- ▶ Buttons/Checkbox
- ▶ Bold/Italic/Color
- ▶ Comments
- ▶ Text Box
- ▶ Email/ Password

Front End



Three Layers of Web Design

Structure	Style	Behavior
HTML markup	CSS	JavaScript



Tags

- ▶ Tags are enclosed in <angle brackets>
- ▶ The closing tag has the forward slash </tag>
- ▶ Tags can be upper or lower case
- ▶ Historically lower case is used and is generally recommended

Attributed

- ▶ Attributes add discretionary properties to tags
- ▶ They are added to the opening tag within the <> brackets
- ▶ For example `<html lang="us-en">` adds a language attribute to the `<html>` tag which is useful for search engines and assistive devices
- ▶ Attributes can use single or double quotes - or no quotes
- ▶ Best practice suggests always quote attributes - and stay consistent with style

Basic HTML Template

```
<html>
<head>
  <title> HTML Template
</title>
</head>
<body>
  <h1> Main Heading Here
</h1>
  <p> Paragraph here. </p>
</body>
</html>
```

- ▶ Copy this code into any text editor
- ▶ Save it as first.html
- ▶ (or give it any other name.html)
- ▶ Double click the file to open it in your browser

Hello World!!

```
<html lang="us-en">
<head>
  <title> Hello, World!
</title>
</head>
<body>
  <h1> Hello, World! </h1>
  <p> My first HTML web
  page.</p>
</body>
</html>
```

- ▶ Change the title and body text from the previous code
- ▶ Add the language attribute
- ▶ Add additional paragraph
- ▶ Save it with different file name filename.html

Headings

```
<body>
<h1> This is Heading 1 </h1>
<h2> This is Heading 2 </h2>
<h3> This is Heading 3 </h3>
<h4> This is Heading 4 </h4>
<h5> This is Heading 5 </h5>
<h6> This is Heading 6 </h6>
</body>
```

- ▶ There are six heading levels <h1> to <h6> each with a smaller font size
- ▶ The levels represent a semantic hierarchy indicating the relative importance of each heading
- ▶ <h1> to <h3> are most common
- ▶ Formatting is the domain of CSS
- ▶ Create a page with different heading levels to see the difference

Summary

Tags	Description
<code><html></code>	Tells the browser this is an HTML5 document
<code></html></code>	Delimits the HTML document
<code>Lang="en-us"</code>	Adds the language attribute
<code><head> </head></code>	Delimits an area for information
<code><title> </title></code>	Creates a title in the browser title bar
<code><body></body></code>	Establishes the content to display
<code><h1></h1></code>	Declares a heading
<code><p> </p></code>	Declares a paragraph

Unordered Lists

```
<ul>
```

```
  <li> Item </li>
```

```
  <li> Item </li>
```

```
  <li> Item </li>
```

```
</ul>
```

► **** tags define a non-sequential bulleted list (an “unordered” list)

► **** tags define a list item within the unordered list

Ordered Lists

`<ol>`

`<li> Item 1 </li>`

`<li> Item 2 </li>`

`<li> Item 3 </li>`

`</ol>`

► `<ol>` tags define a sequential numbered list (an “ordered” list)

► `<li>` tags define a list item within the ordered list

Nested Lists

```
<ul>
```

```
  <li> Item 1 </li>
```

```
  <li> Item 2 </li>
```

```
  <ol>
```

```
    <li> Sub Item 1 </li>
```

```
    <li> Sub Item 2 </li>
```

```
  </ol>
```

```
  <li> Item 3 </li>
```

```
</ul>
```

- ▶ Nested lists create a list within a list
- ▶ Copy the nested code opposite
- ▶ Switch the ordered and unordered lists
- ▶ Add the type attribute to each list

Anchor Tags-Hyperlinks

```
<a href="url"> Link Text  
</a>
```

- ▶ Links are created using the anchor **<a>** **** tags
- ▶ The link address is specified in the **href** attribute
- ▶ **Link Text** becomes the underlined link the user clicks on to activate the link

Anchor Tags-Hyperlinks

```
<a href="http://www.google.com">  
  Google </a>
```

```
<a href="first.html">  
  My First Web Page </a>
```

```
<a href="#top"> Top of Page </a>
```

- ▶ Link to an external site
 - ▶ Add several external links to a page
- ▶ Link to a local file
 - ▶ Create a list of the web pages you saved and add local links (menu)
- ▶ Link to an internal page marker

Image Tag

```

```

- ▶ **** tags have no closing tag
- ▶ Attributes can be written as a single line of text or with whitespace for added readability
- ▶ The browser ignores whitespace
- ▶ The **src** attribute is the image path
- ▶ The **alt** attribute is for replacement
- ▶ Include **width** and **height** for browser

Image as Links

```
<a href="http://www.website.com">  
  
</a>
```

- ▶ Create a clickable image
- ▶ Wrap an `<img>` tag inside an anchor `<a>` tag

Tables

▶<table> tags define a data table

▶<tr> tags define a row with line breaks

<table>

<tr>

<td> Row 1 Col 1</td>

<td> Row 1 Col 2</td>

<td> Row 1 Col 3</td>

<td> Row 1 Col 4</td>

<td> Row 1 Col 5 </td>

</tr>

</table>

▶<td> tags define a data cell, and display inline, horizontally, in a row

▶Copy this code to create a table with 2 rows with 3 columns

▶Add additional rows and columns

Table Headings with Border

```
<table border="1">
  <tr>
    <th> Col 1 Heading </th>
    <th> Col 2 Heading </th>
    <th> Col 3 Heading </th>
  </tr>
  <tr>
    <td> Row 1 Col 1 </td>
    <td> Row 1 Col 2 </td>
    <td> Row 1 Col 3 </td>
  </tr>
</table>
```

- ▶ **<th>** tags are inline elements that can be used to define column headings
- ▶ **<th>** tags replace **<td>** tags in the top row for column headings
- ▶ The first **<td>** tag of create row labels

Caption

```
<table>
```

```
  <caption>
```

```
    <p> Title Goes Here
```

```
    <p> This table shows . . .
```

```
  </caption>
```

```
    ...
```

```
</table>
```

- ▶ The **<caption>** tag is a container for a table title and/or description
- ▶ The **<caption>** tag must directly follow the `<table>` tag
- ▶ Long captions can also be done with standard html tags placed before the table tag

Summary

Tags	Description
<code><table> </table></code>	<table> tag defines a table container
<code><tr> </tr></code>	<tr> tags define a table row
<code><td> </td></code>	<td> tags define a cell for table data
<code><th> </th></code>	<th> tags define a table heading
<code>colspan="2"</code>	colspan attribute defines the number of columns a cell should span.
<code>rowspan="2"</code>	rowspan attribute defines the number of rows a cell should span.
<code><caption> </caption></code>	<caption> provide for a title and brief description

Quotations

<blockquote>

Four score and seven years
ago our fathers brought forth,
upon this continent, a new
nation, conceived in liberty,
and dedicated to the proposition
that all men are created equal.

</blockquote>

- ▶ **<q>** tags are used for short inline quotes with quotation marks
- ▶ He said, **<q>** I will return **</q>**
- ▶ The **<blockquote>** tag is a block level element used for longer passages and displays with indented text

Line Break

`<p>`

Without line break tags `<br>`
the browser would display

`<br>`

this as one continuous
paragraph.

`</p>`

- ▶ Web browsers ignore whitespace
- ▶ The `<br>` tag can be used to force a line break

Comments and Horizontal Rule

`<!-- comment -->`

`<hr>`

- ▶ Add comments to html code where needed
- ▶ The `<hr>` tag adds a horizontal rule as separator line across the screen

Forms

```
<h2>HTML Forms</h2>
```

```
<form action="#top">
```

```
  <label for="fname">First name:</label><br>
```

```
  <input type="text" id="fname" name="fname" value="John"><br>
```

```
  <label for="lname">Last name:</label><br>
```

```
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
```

```
  <input type="submit" value="Submit">
```

```
</form>
```

```
<p>If you click the "Submit" button, the form-data will be sent to a  
page called "/action_page.php".</p>
```

HTML Forms

First name:

Last name:

If you click the "Submit" button, the form-data will be sent to a page called "/action_page.php".

Select

<h2>The select Element</h2>

<p>The select element defines a drop-down list:</p>

<form action="/action_page.php">

 <label for="cars">Choose a car:</label>

 <select id="cars" name="cars">

 <option value="volvo">Volvo</option>

 <option value="saab">Saab</option>

 <option value="fiat">Fiat</option>

 <option value="audi">Audi</option>

 </select>

 <input type="submit">

</form>

The select Element

The select element defines a drop-down list:

Choose a car:

HTML Tags

Tag	Description
<html>	Defines an HTML document
<title>	Defines a title for the document
<body>	Defines the document's body
<h1> to <h6>	Defines HTML headings
<p>	Defines a paragraph
 	Inserts a single line break
<a>	Defines a hyperlink
<!--...-->	Defines a comment
	Defines important text
	Defines an image
<form>	Defines an HTML form for user input

HTML Tags

Tag	Description
<input>	Defines an input control
<textarea>	Defines a multiline input control (text area)
<button>	Defines a clickable button
<select>	Defines a drop-down list
<optgroup>	Defines a group of related options in a drop-down list
<option>	Defines an option in a drop-down list
<label>	Defines a label for an <input> element
<fieldset>	Groups related elements in a form

Text Formatting

- ▶ CSS is the preferred way to change the browser's display style
- ▶ The separation of style and markup is a recommended best practice
- ▶ However, there are some basic text formatting tags that can be used in HTML
- ▶ These are inline tags used to mark text within the flow of another tag
- ▶ Those included here are generic tags - without any semantic meaning

CSS Linking Styles

- ▶ **Inline** - by using the style attribute inside HTML elements

```
<h1 style="color:blue;"> This is Heading </h1>  
<p style="color:red;"> write a paragraph</p>
```

- ▶ **Internal** - by using a <style> element in the html file

```
<html>  
  <head>  
    <style>  
      body {background-color: powderblue;}  
      h1   {color: blue;}  
      p    {color: red;}  
    </style>  
  </head>  
  <body>  
    <h1>This is a heading</h1>  
    <p>This is a paragraph.</p>  
  </body>  
</html>
```

CSS Linking Styles

- ▶ CSS Linking styles
- ▶ **External** - by using a <link> element to link to an external CSS file

```
<html>
```

```
  <head>
```

```
    <link rel= "stylesheet" href="file.css">
```

```
  </head>
```

```
  <body>
```

```
    <h1>This is a heading </h1>
```

```
    <p>This is a paragraph. </p>
```

```
  </body>
```

```
</html>
```

Some Examples of solving HTML

```
<center>
  <div class="">
    <p class="contact-info">
      <span class="contact-icon"><i class="fas fa-phone"></i></span>
(123) 456-7890
<span class="contact-icon">
<i class="far fa-envelope"></i></span>
john.doe@email.com <span class="contact-icon"><i class="fas fa-
globe"></i></span>
<a href="https://www.linkedin.com/in/johndoe" target="_blank
LinkedIn Profile</a>

</p>
</div>
</center>

<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/5.15.3/css/all.min.css"> <!-- Include Font Awesome for
icons -->
```

```
<div class="">
<h1> <span style="background-color:black;color:white;">
Professional Profile</span> </h1>

<p> add text here for professional profile</p>

</div>
```

```
<div >
<h1><span style="background-color:black;color:white;">Core
Skills</span> </h1>
```

```
<ul type="square" class="parallel-ul ">
  <li>C++</li>
  <li>Python</li>
  <li>C</li>
</ul>
<ul class="parallel-u0 ">
  <li>C#</li>
  <li>Java</li>
  <li>OOP</li>
</ul>

</div>
```

```
.parallel-ul {
    float: left;
    width: 50%;

}
.parallel-u0 {
    float: right;
    width: 50%;
padding: 0;    }
```

Some Examples of solving HTML

```
.container {  
    background-color: #fff;  
    padding: 10px;  
    border-radius: 2px;  
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);  
    overflow: hidden; /* Clear floating elements inside  
the container */  
}
```

```
<div class="">  
<h1><span style="background-color:black;color:white;">Educations  
</span> </h1>  
<ul class="parallel-u0 ">  
    <li><b>MSCS</b> From FAST </li>  
    <p>PHD <br></br> Continue ... </p>  
  
</div>
```

```
<div class="container">  
<h2 >profesional</h2>  
  
</div>
```

Link Of Udemy Course

- ▶ You are required to cover first three sections of this course
- ▶ You have to implement all the content covered in the videos
- ▶ Deadline □ Sunday 11:59pm
- ▶ Note that Viva will also be conducted

- ▶ Link:

<https://www.udemy.com/course/design-and-develop-a-killer-website-with-html5-and-css3/?couponCode=ST8MT101424>